BUSINESS REPORT

Telecom Customer Churn Analysis

Understanding Customer Behavior and Identifying Opportunities to Improve Retention

7,043
Customer Records

SQL Server
Data Analysis

Google Sheets
Data Visualization



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Business Problem & Objective

Understanding why customers leave and how to improve retention

Business Problem

Customer retention is a major challenge in the telecom industry because losing customers directly impacts revenue. The business wanted to understand why customers are leaving and identify the customer segments most likely to churn.

Objective

My objective was to analyze customer data, identify the key drivers of churn, and provide recommendations to improve customer retention.

Key Questions

- What is the overall churn rate?
- Which contract types have higher churn?
- Which internet services are associated with higher churn?
- Which payment methods are linked to higher churn?





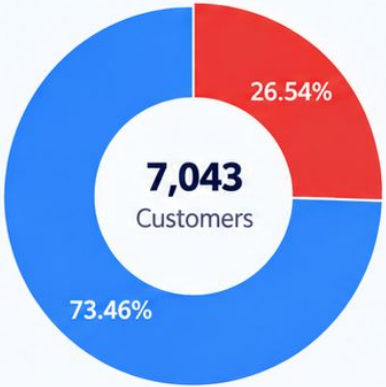
Dataset Overview & Key Metrics

A snapshot of the customer data and overall churn performance.

Total Customers	Churned Customers	Retained Customers	Overall Churn Rate
7,043	1,869	5,174	26.54%

Customer Distribution

Proportion of churned and retained customers



- Churned Customers
  1,869 (26.54%)
- Retained Customers
  5,174 (73.46%)

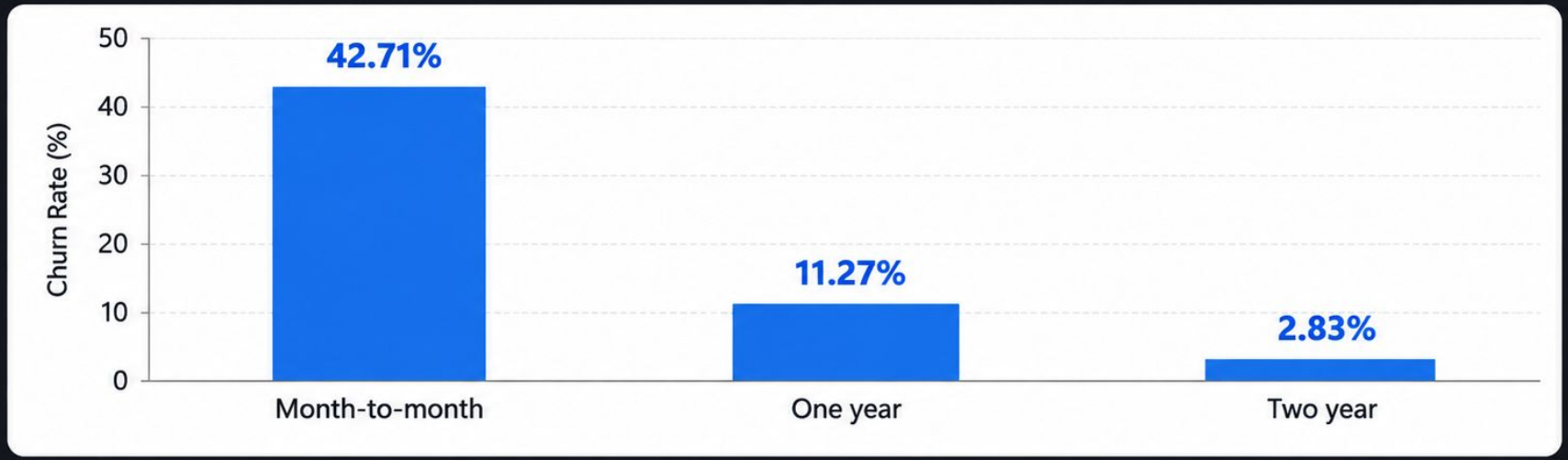
Dataset Details

Source	Kaggle – Telco Customer Churn Dataset
Total Records	7,043 customer records
Total Columns	11 (selected for analysis)
Key Columns	CustomerID, gender, SeniorCitizen, Partner, Dependents, tenure, InternetService, Contract, PaymentMethod, MonthlyCharges, TotalCharges, Churn

Analysis & Visualizations

Contract Type Analysis

CHURN RATE BY CONTRACT TYPE

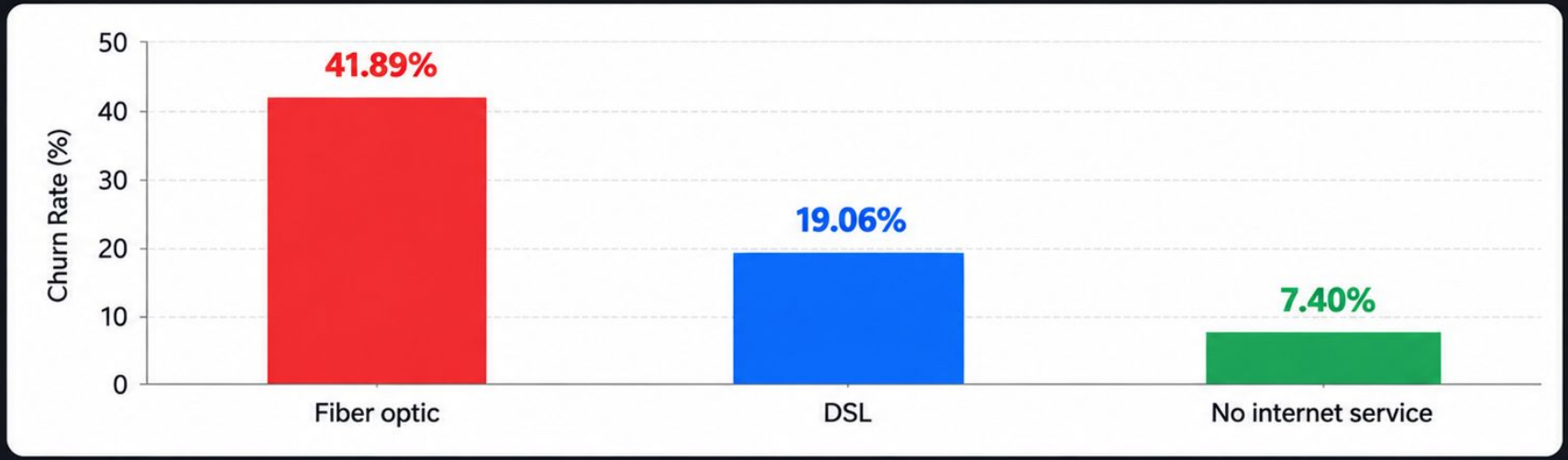


Insight: Month-to-month customers had the highest churn rate (42.71%), nearly 15 times the two-year contract rate (2.83%). Prioritize this group for retention offers.

Analysis & Visualizations

Internet Service Analysis

CHURN RATE BY INTERNET SERVICE

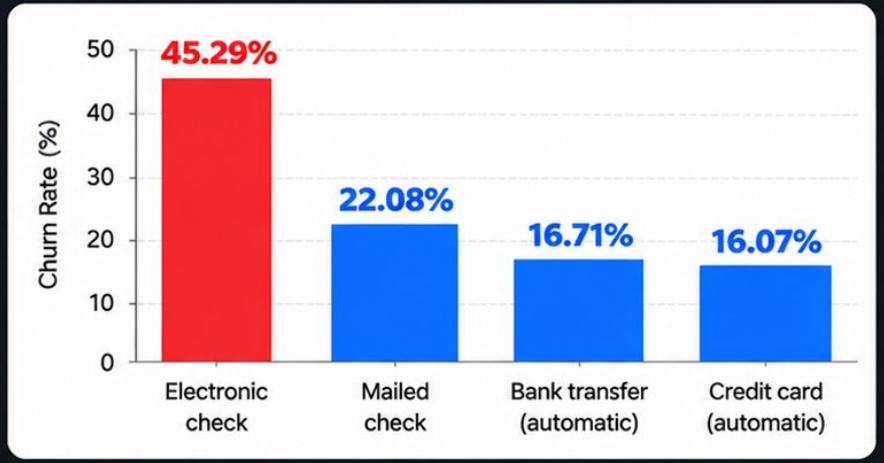


Insight: Fiber optic customers had the highest churn rate (41.89%), significantly higher than DSL (19.06%) and customers with no internet service (7.40%). This indicates that fiber optic customers are a high-risk segment and the business should investigate service quality, pricing and customer support to reduce churn.

Analysis & Visualizations

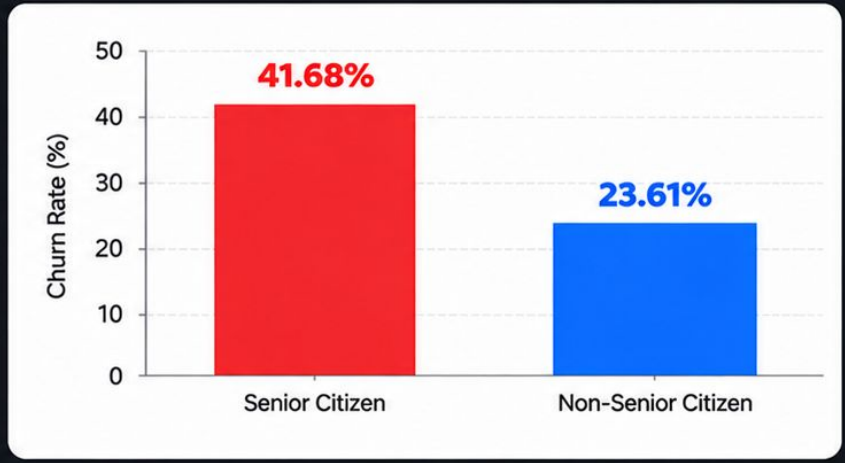
Payment Method & Senior Citizen Analysis

CHURN RATE BY PAYMENT METHOD



Insight: Customers using electronic check have the highest churn rate **(45.29%)**, significantly higher than other payment methods. This suggests a need to review the billing experience or encourage customers to switch to automatic payment methods.

CHURN RATE BY SENIOR CITIZEN STATUS

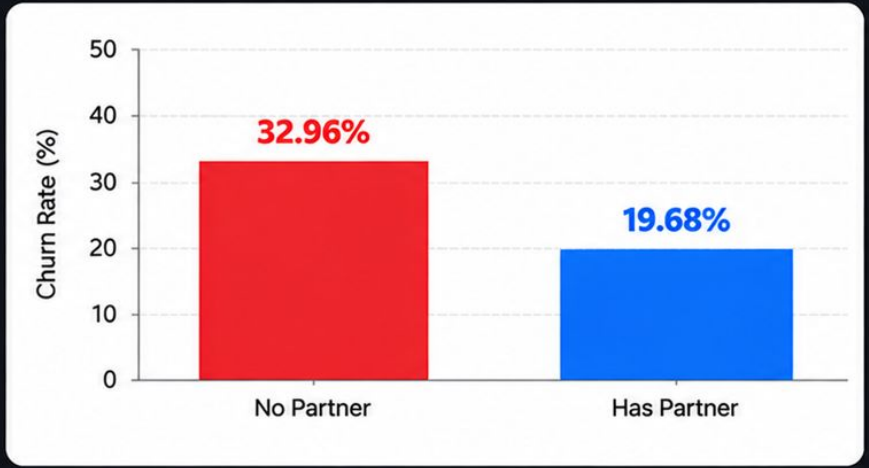


Insight: Senior citizens have a churn rate of **41.68%**, which is significantly higher than non-senior citizens **(23.61%)**. This indicates that senior citizens are a higher-risk customer segment and may require targeted support and retention strategies.

Analysis & Visualizations

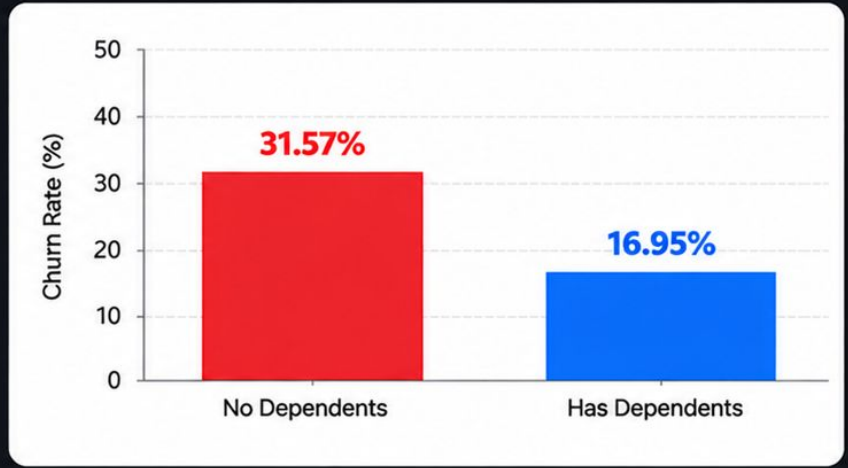
Partner & Dependents Analysis

CHURN RATE BY PARTNER STATUS



Insight: Customers without a partner have a churn rate of **32.96%**, which is significantly higher than customers with a partner **(19.68%)**. This suggests that customers without a partner may require more engagement and personalized offers.

CHURN RATE BY DEPENDENTS STATUS



Insight: Customers without dependents have a churn rate of **31.57%**, which is significantly higher than customers with dependents **(16.95%)**. This indicates that customers without dependents are a higher-risk segment and may need targeted retention strategies.

STRATEGIC ROADMAP

Strategic Recommendations

QUICK WINS (0–3 Months)

- Prioritize month-to-month customers with targeted retention offers.
- Review billing friction among electronic-check users and test automatic-payment options.
- Investigate fiber optic customer complaints, service quality, and pricing.
- Contact new customers during their first six months with proactive onboarding support.

LONG-TERM (3–12 Months)

- Test incentives that encourage month-to-month customers to move to annual contracts.
- Use churn trends to identify high-risk customer segments and guide retention campaigns.
- Monitor churn by contract, internet service, payment method, tenure, and monthly charges.
- Measure campaign impact against a comparison group and track cost per retained customer.

PROJECT SUMMARY

Final Thoughts

PROJECT SUMMARY

Objective
Analyze customer data to identify key churn drivers and provide actionable recommendations to improve customer retention.

Dataset
Telco Customer Churn Dataset (7,043 customers) with 21+ features.

Tools Used
SQL Server, Google Sheets, Pivot Tables, Charts.

Key Findings
Highest churn in month-to-month contracts (42.71%), fiber optic (41.89%), electronic check (45.29%), and customers without partner/dependents.

Business Impact
Helps the business focus retention efforts on high-risk segments and reduce revenue loss.

NEXT STEPS

Deep Dive Analysis
Validate findings with additional data and explore other potential factors.

Testing Recommendations
Run A/B tests for contract, pricing, and payment strategies to measure real impact.

Build Predictive Model
Develop a churn prediction model to identify at-risk customers in advance.

Monitor & Optimize
Track key metrics, review performance regularly, and refine strategies based on results.

Explore Further Opportunities
Analyze additional segments like customer satisfaction, usage patterns, and product mix for deeper insights.